PHYS7101: NUMERICAL METHODS AND DATA ANALYSIS · LECTURE 8

Updating what we know

Preparing and relating measured signals;
Bayesian inference and Markov chain Monte Carlo

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Where we left off

From earlier lectures

- Interpolation (L4); MAD, bootstrap, surrogates (L5)
- Maximum likelihood; Poisson counts of radioactive decay, $\tau = 20\,\mathrm{s}$ (L3)
- A distribution family only when physics supplies it (L5)
- Monte Carlo (L2, L5); autocorrelation (L6); correlation (L7)

Today

A Two measured signals: preparing and relating them

B Bayes' theorem: likelihood, prior, posterior; credible intervals

C Sampling the posterior: Metropolis–Hastings

D Outlook: parameters that change in time

Part A uses no model of the data; Parts B–D do. Bayesian inference needs a model of how the data arise: use it when physics supplies that model – as for a parametric description in Lecture 5.

PART A

Two measured signals

Heartbeat and breathing: preparing the data, relating the signals, and the size of a change – without a model of the data.

Where the data come from

Problem: the signals of Parts A and D are simulated, so that the truth is known. What produces them?

Two phases, breathing φ_r and heart φ_h:

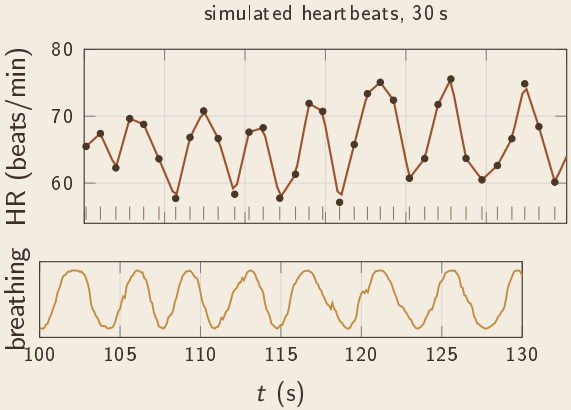
$$\dot{\varphi}_r = 2\pi \times 0.25\ \text{Hz} + \xi_r(t)$$
$$\dot{\varphi}_h = 2\pi \times 1.1\ \text{Hz} + \varepsilon \sin \varphi_r + \xi_h(t)$$

- ε: coupling (rad/s) – breathing speeds the heart up and slows it down
- ξ_r, ξ_h: independent Gaussian noise
- solved with Euler–Maruyama (L2), step 0.01 s
- breathing signal: $\cos \varphi_r$; a **heartbeat** each time φ_h completes a cycle

Slides	Data
5–7	one 300-s recording, $\varepsilon = 0.8$ rad/s; artefacts added by hand (one beat deleted, one false beat inserted); "another recording": same model, different noise
8	24 recordings, one per "person": 12 lying (ε random in 0.6–1.2 rad/s), 12 others standing (ε also reduced by a random factor 0.5–0.9)
25	1000 s with ε changing in time

From event times to an evenly sampled series

Problem: a heartbeat is an event in time, not a value at regular times. How do we get a series for the methods of Lectures 6 and 7?



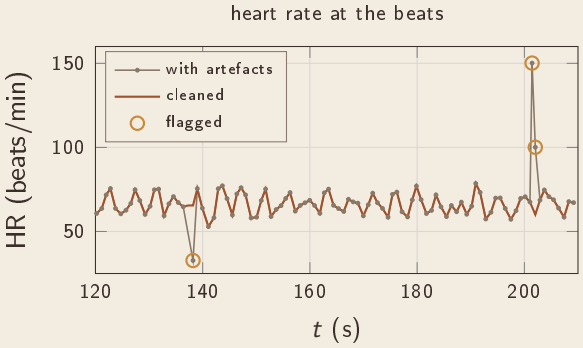
1. beat times t_j (ticks)
2. intervals $\mathrm{RR}_j = t_j - t_{j-1}$
3. heart rate $60/\mathrm{RR}_j$ (beats/min) at t_j: **uneven** times (dots)
4. interpolate (L4) to an even grid: here 4 Hz, cubic spline (line)

Caution: the beats sample the breathing rhythm only ≈ 4.4 times per breath. Linear interpolation flattens it (L4): standard deviation 4.6 bpm, against 5.3 with the spline and 5.5 at the beats.

Data: heart–breathing model of slide 4, $\varepsilon = 0.8$ rad/s, 300 s.

Artefact cleaning

Problem: one beat missed by the detector, one false extra beat. What do they do, and how do we find them?



Effect: a missed beat doubles one interval (heart rate 33 bpm); an extra beat splits one (150 bpm). One such value dominates a mean, a standard deviation or a spectrum.

Find (nonparametric, L5): compare each interval with the median of its 11 neighbours; flag it if the difference exceeds 5 MAD. Here MAD = 0.044 s: 3 intervals flagged.

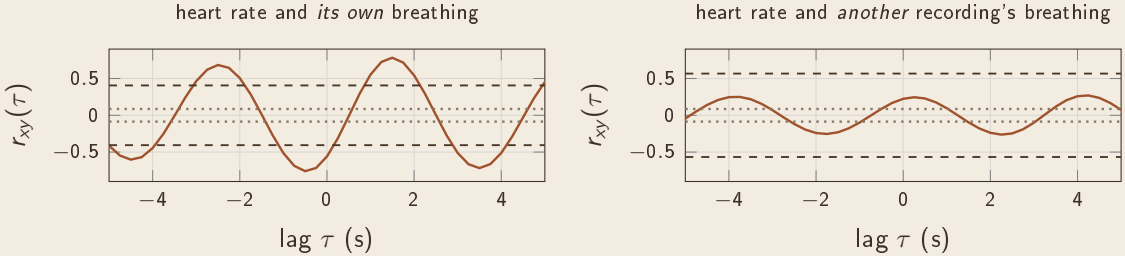
Correct: a doubled interval → insert a beat in the middle; two short ones → remove the beat between them.

Report: 1 beat inserted, 1 removed, of 329.

Data: as on slide 5; one beat deleted (138 s) and one false beat inserted (201 s) by hand.

Cross-correlation, tested with surrogates

Problem: does breathing influence the heart rate, and with what delay? Is the answer more than chance?

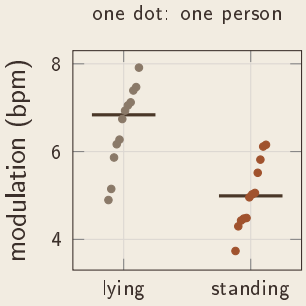


Cross-correlation $r_{xy}(\tau)$: the correlation (L7) of $x(t)$ with $y(t-\tau)$; for $y = x$ the autocorrelation (L6). Left: 0.78 at $\tau = 1.5$ s – the heart rate follows breathing. 95% levels of the largest $|r_{xy}|$: **shuffled** heart rate (dotted, L5) destroys its memory, level 0.09 – even the unrelated pair (0.27) looks "significant"; **time-shifted** breathing (dashed) keeps each signal's memory – left significant, right not (approximately, for such regular rhythms).

Data: slide 4: the cleaned recording of slide 6; right, breathing from a second run with different noise.

Effect sizes with bootstrap intervals

Problem: heart-rate modulation in 12 people lying and 12 *other* people standing. How large is the difference, and how sure are we?



Effect size: how large the difference is, in units we can read – not only whether it exists.

effect size	value	95% interval
difference standing − lying (Hodges–Lehmann, L5)	−1.68 bpm	−2.42 to −0.85
P(standing < lying) for a random pair	0.91	0.77 to 1.00

Bootstrap percentile interval (L5): draw 12 values at random from each group, with repeats; recompute; repeat 20 000 times; take the 2.5% and 97.5% points. Resample people, who are independent – not samples within one recording, which are not.

Data: 24 runs of the model of slide 4, one per "person".

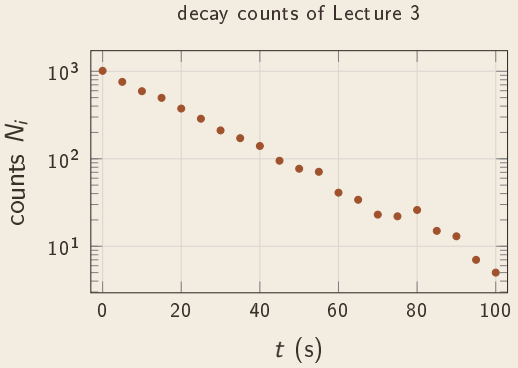
PART B

Bayes' theorem

Likelihood, prior, posterior – for one parameter.

Two questions about the same data

Problem: after measuring the counts, what do we know about the decay time τ, and how sure are we?



Model: $N = N_0 e^{-t/\tau}$, counts every 5 s.

Lecture 3: which τ makes the data most probable?
One number: the best fit.

Today: how probable is each τ, given the data?
A distribution: what we know about τ.

For now $N_0 = 1000$ is taken as known; both parameters in Part C.

Data: decay counts simulated in Lecture 3: $N_0 = 1000$, $\tau = 20$ s, Poisson noise.

Bayes' theorem

Problem: how do we turn "probability of the data given θ" into "probability of θ given the data"?

Both orders of the joint probability are equal: $p(\theta, d) = p(d \mid \theta)\, p(\theta) = p(\theta \mid d)\, p(d)$. Hence

$$\boxed{p(\theta \mid d) = \frac{p(d \mid \theta)\; p(\theta)}{p(d)}}$$

$p(\theta \mid d)$	**posterior**	what we know about θ after the data d
$p(d \mid \theta)$	**likelihood**	how probable the data are for each θ: the model of the data
$p(\theta)$	**prior**	what we know about θ before the data
$p(d)$	**evidence**	a number that makes the posterior integrate to 1

In words: posterior $\propto$ likelihood $\times$ prior.

The likelihood: the model of the counts

Problem: how probable are the observed counts for a given τ?

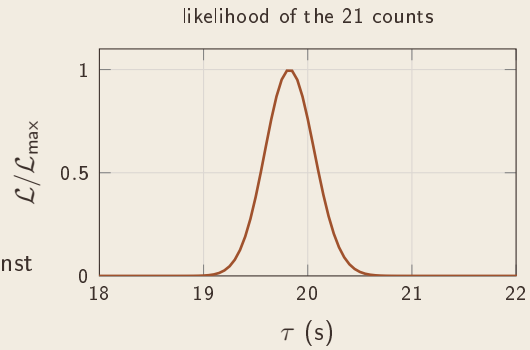
Expected count at t_i: $\mu_i = N_0 e^{-t_i/\tau}$. Counts of independent decays are Poisson (L5):

$$p(N_i \,|\, \tau) = \frac{\mu_i^{N_i} e^{-\mu_i}}{N_i!}$$

Independent counts: multiply,

$$\mathcal{L}(\tau) = \prod_i p(N_i \,|\, \tau), \quad \ln \mathcal{L} = \sum_i (N_i \ln \mu_i - \mu_i) + \text{const}$$

Its maximum: $\hat{\tau} = 19.83\,\text{s}$, the **maximum-likelihood** value (L3).



For Gaussian errors, $-2 \ln \mathcal{L} = \chi^2 + \text{const}$: least squares (L3) is the Gaussian case.

The prior: what we knew before

Problem: the likelihood is not a distribution of τ. What else does Bayes' theorem need?

The **prior** $p(\tau)$ states, as a distribution, what we know before these counts:

Prior	Knowledge it encodes	Here
flat	only physical limits	$\tau > 0$ (and below 100 s)
informative	an earlier measurement or theory	an earlier result $\tau = 24 \pm 1.5$ s, Gaussian

The prior is part of the result

State it, and check how much the result depends on it (slide 16). Prior knowledge enters here – openly – and nowhere else.

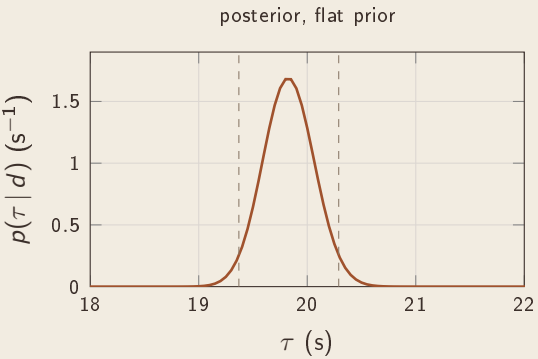
The posterior on a grid

Problem: how do we compute the posterior in practice?

On a grid of τ values (here 2201, from 15 to 26 s):

1. compute $\ln \mathcal{L} + \ln p(\tau)$
2. subtract the maximum, exponentiate
3. divide by the area: $\int p(\tau \mid d)\, d\tau = 1$

	median	19.83 s
Flat prior:	68% interval	19.59–20.06 s
	95% interval	19.37–20.29 s



Dashed: 95% interval; 2.5% of the area lies beyond each line.

Data: decay counts of Lecture 3.

Credible and confidence intervals

Problem: what does "95%" mean for the interval 19.37–20.29 s?

Credible interval (Bayesian)

Given these data and this prior, τ lies in the interval with probability 95%. A statement about τ, for *this* data set.

Confidence interval (frequentist)

A procedure that, over many repeated experiments, gives intervals containing the true τ in 95% of them. A statement about the procedure.

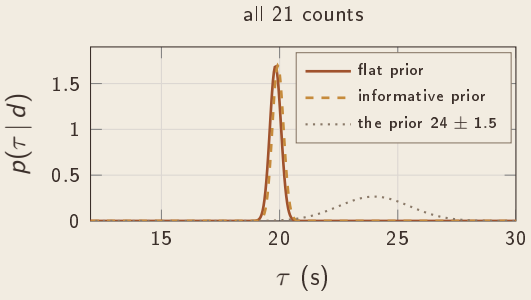
Check by simulation: 2000 repeated experiments with $\tau = 20$ s; the flat-prior 95% credible interval contained 20 s in 95.9% of them. Here the two agree.

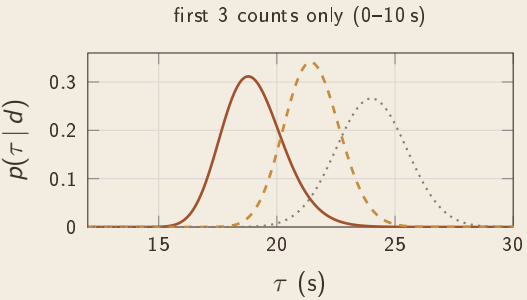
They need not agree when the data are few or the prior is informative – next slide.

Data: 2000 new sets of counts simulated as in Lecture 3 ($\tau = 20$ s).

How much does the prior matter?

Problem: would a different prior change the conclusion?





median, 95% interval	flat prior	prior 24 ± 1.5 s
all 21 counts	19.83, 19.37–20.29	19.93, 19.47–20.39
first 3 counts	18.95, 16.69–21.83	21.48, 19.29–23.87

Many data: the data decide. Few data: the prior decides – here towards a wrong value.

Data: decay counts of Lecture 3; right, only the first three.

PART C

Sampling the posterior

When a grid is too costly: Markov chain Monte Carlo.

Why a grid fails in many dimensions

Problem: a model with d parameters. How many posterior evaluations does a grid need?

parameters d	grid points, 100 per parameter	at 10^6 evaluations per second
1	10^2	instant
2	10^4	instant
5	10^{10}	3 hours
10	10^{20}	3 million years

The cost grows as 100^d (complexity, L1). And most grid points lie where the posterior is almost zero.

Idea

Instead of evaluating everywhere, draw **samples** from the posterior: values of $\boldsymbol{\theta}$ that appear as often as their probability. Then every summary – median, interval, correlation – is computed from the samples, as in Monte Carlo (L2, L5).

The Metropolis–Hastings algorithm

Problem: how do we draw samples from a posterior we can evaluate only up to a constant?

Start at $\boldsymbol{\theta}_0$. At each step:

1. **propose** $\boldsymbol{\theta}' = \boldsymbol{\theta}_n +$ a random step (Gaussian, width s)
2. **accept** with probability

$$\min\left(1,\ \frac{p(\boldsymbol{\theta}' \mid d)}{p(\boldsymbol{\theta}_n \mid d)}\right)$$

3. if accepted, $\boldsymbol{\theta}_{n+1} = \boldsymbol{\theta}'$; if not, $\boldsymbol{\theta}_{n+1} = \boldsymbol{\theta}_n$ (counted again)

Why it works

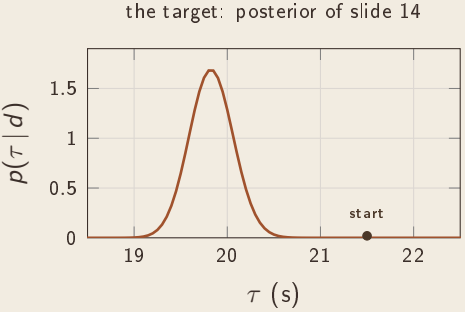
Uphill moves are always accepted, downhill moves sometimes. In the long run each $\boldsymbol{\theta}$ is visited in proportion to its posterior probability.

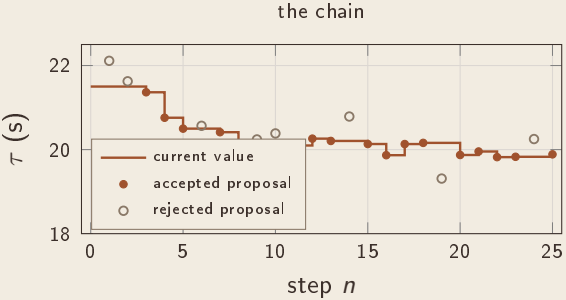
Only the **ratio** is needed: the evidence $p(d)$ cancels.

The samples form a **Markov chain**: each depends only on the one before. Successive samples are therefore not independent.

Metropolis–Hastings step by step

Problem: what does the chain actually do? The first 25 steps for τ alone ($N_0 = 1000$), steps of width 0.3 s.





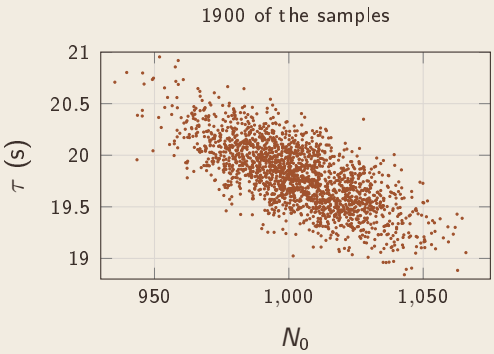
Start at 21.5 s, in the tail. Proposals uphill (towards 19.8 s) are always accepted; downhill ones sometimes. A rejected proposal (open circle) leaves the chain where it is, and that value is counted again. 16 of 25 proposals accepted; after about 20 steps the chain moves around the peak, visiting values as often as the posterior says.

Data: decay counts of Lecture 3.

Metropolis–Hastings in action: N_0 and τ

Problem: both N_0 and τ unknown (flat priors, both positive), as in Lecture 3.





Start far away, $(N_0, \tau) = (500, 60\,\text{s})$; steps of width 15 and 0.4 s; 20 000 steps, 46% accepted. The chain reaches the posterior after about 350 steps: the first 1000 (**burn-in**, before the dashed line) are discarded. Right: the samples form the tilted cloud of a correlated posterior (L3).

Data: decay counts of Lecture 3.

Checks: step width and burn-in

Problem: how do we choose the step width, and how do we know the chain has left its start?



Step width: tiny steps are almost all accepted, but the chain creeps; huge steps are almost all rejected, and it stands still. Both explore the posterior slowly. A width that accepts a sizeable fraction – often 20–50% – moves well.

Burn-in: the trace of slide 21 drifts from its start (60 s) for about 350 steps. Discard such a stretch; here the first 1000 steps.

Look at the trace: it should wander around one level, with no trend.

Data: decay counts of Lecture 3.

Results from the samples

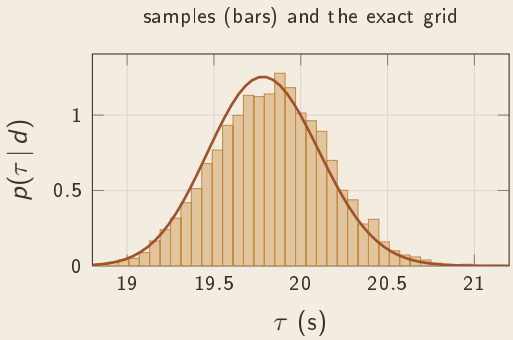
Problem: what do we report, and is the sampler right?

From the 19 000 samples:

	median	95% interval
τ (s)	19.81	19.17–20.44
N_0	1002	963–1045

Correlation of N_0 and τ: -0.70.

Standard deviation of τ: 0.33 s – the ± 0.34 s spread of the weighted fit in Lecture 3.



Check: on a 2-D grid, τ 19.79, 19.18–20.42 s.

With two parameters the grid is still possible – and confirms the sampler. With ten, only the sampler remains: test it first where the answer is known.

Data: decay counts of Lecture 3.

PART D

Outlook

When the parameters themselves change in time.

Dynamical Bayesian inference

Problem: two oscillators – like breathing and heartbeat – whose coupling changes in time. Can we follow the coupling from the two phases?

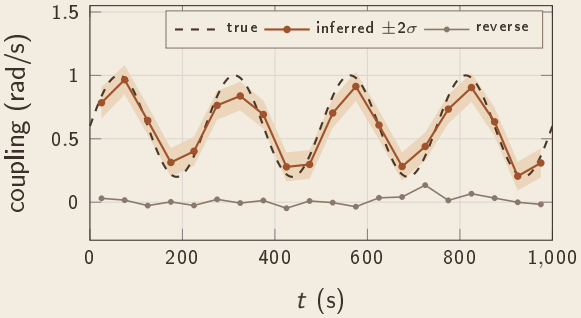
Model: for the faster oscillator, phase φ_h,

$$\dot{\varphi}_h = \sum_{k=1}^{7} c_k \, \Phi_k(\varphi_h, \varphi_r) + \xi_h(t)$$

$\Phi_1 = 1$ (so c_1 is the frequency ω_h), and sine and cosine of φ_h, φ_r, $\varphi_h - \varphi_r$; ξ_h: Gaussian noise.

Inference in windows of 50 s: the equation is linear in the c_k and the noise Gaussian, so the posterior is Gaussian and exact, as for least squares (L3). Each window's posterior, widened, is the prior of the next.

simulated: respiration → heart



Inferred coupling follows $\varepsilon(t)$ (90% of windows within $\pm 2\sigma$); reverse: zero within $\pm 2\sigma$ (95%).

Data: model of slide 4, 1000 s, $\varepsilon(t) = 0.6 + 0.4\sin(2\pi t/250\,\mathrm{s})$.

What dynamical Bayesian inference adds

Problem: what does this give that a single fit does not, and what does it assume?

It gives

- parameters **followed in time**: a nonautonomous system (L2) described as such, not as noise
- the **direction** of coupling – within the assumed model: which oscillator drives which
- frequencies and noise strength, each with its uncertainty

It assumes

- a model: phase equations with the chosen functions Φ_k
- Gaussian white noise
- parameters nearly constant within one window

Whole record at once: coupling 0.58 rad/s – an average that hides a fivefold change. Time variability, not noise, once more.

Summary

Step	Tools	Check
Event times to a series	intervals, rate, interpolation to an even grid	the interpolant (cubic, not linear)
Artefacts	running median; flag beyond k MAD	what is flagged; report corrections
Relation of two signals	cross-correlation; largest $\lvert r_{xy} \rvert$	surrogates that keep each signal's memory (time shift)
Size of a change	effect sizes with bootstrap percentile intervals	resample independent units
Model the data	likelihood (Poisson, Gaussian $\leftrightarrow \chi^2$)	does physics supply it?
State prior knowledge	flat or informative prior	repeat with another prior
Posterior, few parameters	grid; median and credible interval	coverage by simulation
Posterior, many parameters	Metropolis–Hastings	trace; burn-in; acceptance; a case with a known answer
Parameters in time	dynamical Bayesian inference	window length; model

Before you report: six questions

1. Did my preparation change the data – interpolant, artefact corrections – and did I report it? *(Part A)*
2. Do my surrogates and resamples keep what H_0 does not question? *(time shift; independent units)*
3. Does physics supply the model of the data? *(otherwise: nonparametric methods, L5)*
4. What is my prior, and does the result change with another one? *(few data: the prior decides)*
5. Has my chain forgotten its start, and does it move well? *(trace; burn-in; acceptance)*
6. Does the sampler reproduce a case with a known answer? *(verification)*

Report the posterior: median, interval, correlations – and the prior.

TO TAKE AWAY

Posterior $\propto$ likelihood $\times$ prior.

Before any model: prepare the data openly, test relations with surrogates that keep each signal's memory, and report the size of an effect with its interval.

With a model that physics supplies: the result is a distribution, not a number – sample it when a grid is too costly, and check the sampler where the answer is known.